Informatica®

Informatica® Cloud Data Integration

Azure AI Search Connector

Informatica Cloud Data Integration Azure AI Search Connector
October 2025

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Preface

Use Azure AI Search Connector to learn how to write to Azure AI Search database by using Data Integration. Learn to create a Azure AI Search connection, run mappings in advanced mode and tasks in Data Integration.

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CHAPTER 1

Introduction to Azure AI Search Connector

You can use Azure AI Search Connector to write unstructured text files, such as PDFs, as vector embeddings to the Azure AI Search vector database.

You can use an Azure AI Search connection as a target in mappings in advanced mode. When you run a mapping in advanced mode or a task based on a mapping in advanced mode, Data Integration analyzes the data logic in the mapping to automatically assign data logic to run on an advanced cluster.

Example

You run a legal firm that handles a massive repository of legal documents, contracts, and case files in PDF format in Microsoft Azure Data Lake Storage Gen2.

Legal professionals need efficient ways to search and retrieve relevant documents quickly, often using complex queries that go beyond simple keyword searches.

To enhance the speed and accuracy of document retrieval, you want to integrate your data into the Azure AI Search vector database. This integration allows legal professionals to efficiently query high-dimensional vectors that represent complex data, such as text and tables, streamlining their search processes and enhancing overall productivity.

To integrate data in Azure AI Search, you perform the following steps:

1. Create a mapping in advanced mode.
2. Add a Source transformation to read data from Microsoft Azure Data Lake Storage Gen2.
3. Add a Chunking transformation to split the text into smaller manageable chunks for enhanced vector representation.
4. Add an Embedding transformation to convert the text to vector embeddings.
5. Add a Target transformation to use the Azure AI Search connection and load the vector embeddings to Azure AI Search.
6. Run the mapping in advanced mode.

Azure AI Search Connector assets

Create assets in Data Integration to integrate data using Azure AI Search Connector.

When you use Azure AI Search Connector, you can include the following Data Integration assets:

- Mappings in advanced mode
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

CHAPTER 2

Connections for Azure AI Search

Create an Azure AI Search connection to securely write data to the Azure AI Search vector database.

You can use an Azure AI Search connection to specify targets in mappings in advanced mode and mapping tasks.

Prerequisites

Before you configure the connection properties, make sure to complete the following prerequisites:

1. Create an Azure AI Search service in the Azure portal.
2. Generate the API key of your Azure AI Search service to authenticate access to the Azure AI Search APIs. For more information on how to create a service and generate an API key, see Create an Azure AI Search service.

Connect to Azure AI Search

Let's configure the Azure AI Search connection properties for Azure AI Search.

Before you begin

Before you get started, you'll need to get the API key from your Azure AI Search service account.

Check out "Prerequisites" on page 8 to learn about the authentication requirements before you configure a connection.

Connection details

The following table describes the basic connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure agent or serverless runtime environment. Hosted Agent is not applicable for mappings in advanced mode. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.
Host Name	The host name of the Azure search service that you want to connect to. For example, https://rnddev.search.windows.net
API Key	The admin API key to authenticate Azure AI Search connection requests. For example, yavMPJHTY11p08JTPkM3NW1ImhJraYycrLAzSeCEQQ5j

CHAPTER 3

Mappings for Azure AI Search

You can configure a mapping in advanced mode to write to Azure AI Search.

Use the Data Integration Mapping Designer to define and configure a mapping in advanced mode for Azure AI Search target. In advanced mode, the Mapping Designer updates the mapping canvas to include transformations and functions that enable advanced functionality.

You can configure a Target transformation to represent a Azure AI Search index. Add the Target transformations in the mapping canvas and configure the Azure AI Search target properties. When you run a mapping in advanced mode, Data Integration analyzes the data logic defined in the mapping to automatically assign data logic to run on an advanced cluster.

Targets for Azure AI Search

Add a Target transformation to write data to a target.

When you add a Target transformation to a mapping in advanced mode, you define the target connection, target objects, and target properties related to the Azure AI Search connection type. You cannot parameterize an Azure AI Search mapping.

The following table describes the Azure AI Search target properties that you can configure in a Target transformation:

Property	Description
Connection	Name of the target connection.
Target Type	Target type. Select **Single Object**.
Object	Name of the target object that represents an index in the Azure AI Search project.
Operation	Target operation. You can use only the upsert operation.
Update Columns	The ID field of the Azure AI Search index that is used as the primary key column when you upsert target data.

The following table describes the advanced properties:

Property	Description
Batch Size	The number of vectors that you can upsert into the Azure AI Search index in a single API request. Default is 100. Maximum value is 1000.

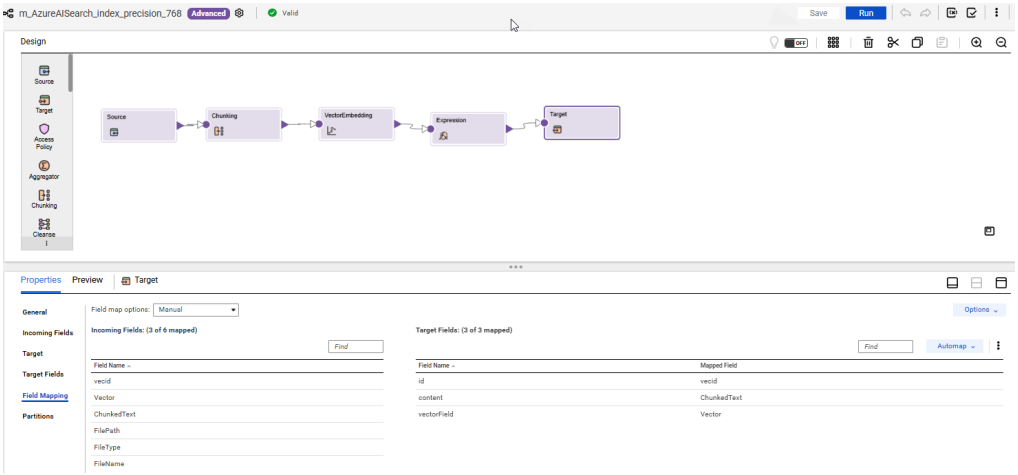
For more information about configuring transformations, see *Transformations* in the Data Integration documentation.

Field mapping

When you configure a Azure AI Search target, you can map the target fields in Azure AI Search to the appropriate incoming fields from the upstream transformation.

To configure the field mapping, perform the following steps:

1. Go to the **Field Mapping** tab.
2. To map the fields, under **Mapped Field**, select the appropriate incoming fields from the list.



Note: When you configure the field mapping, ensure that the precision of the target vector field matches the precision of the incoming vector field.

CHAPTER 4

Data type reference

Data Integration uses the following data types in Azure AI Search mappings in advanced mode:

The following table lists the Azure AI Search data types that are compatible with Data Integration and the corresponding transformation data types:

- Azure AI Search native data types that are inherent to the Azure AI Search database.
- Transformation data types. Set of data types that appear in the transformations. These are internal data types based on ANSI SQL-92 generic data types, which the agent uses to move data across platforms. They appear in all transformations in a mapping in advanced mode.

When the agent writes to a Azure AI Search target, it converts the transformation data types to the comparable native data types.

Azure AI Search and transformation data types

The following table lists the Azure AI Search data types that Data Integration supports and the corresponding transformation data types:

Azure AI Search data type	Transformation data type	Range and Description
Edm.Boolean	Integer	1 or 0
Edm.Double	Double	Precision is 15.
Edm.Int32	Integer	-2,147,483,648 to +2,147,483,647
Edm.Int64	Bigint	-9,223,372,036,854,775,808 to +9,223,372,036,854,775,807 8-byte signed integer.
Edm.String	String	Text data.
Collection (Edm.Single)	Vector	A dense vector with a 32-bit floating point.

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